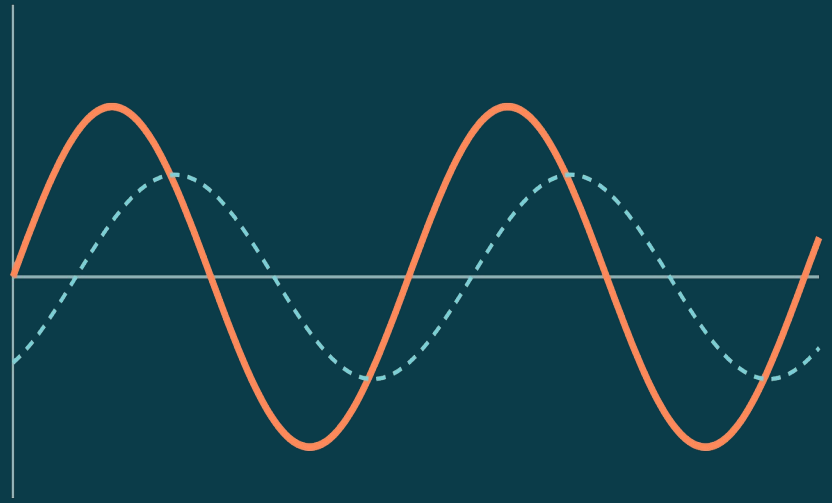
Op-Amp Basics and Amplifier Circuits



Basic Electrical and Electronics Engineering

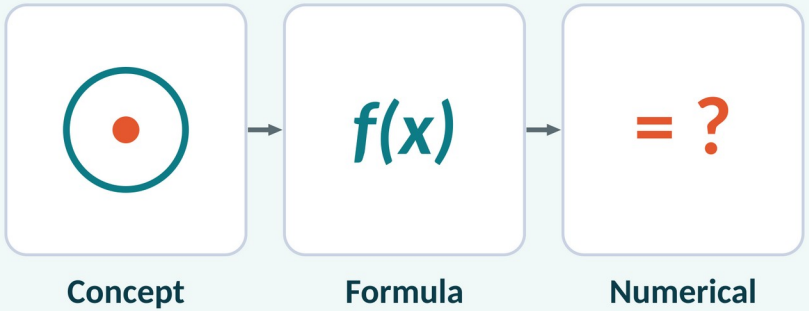
Unit 5: Op-Amp and Linear IC Applications

Topic: Op-amp specifications, differential and common-mode operation, inverting, non-inverting and summing amplifiers

Prof. Mitesh Astik

Learning Outcomes

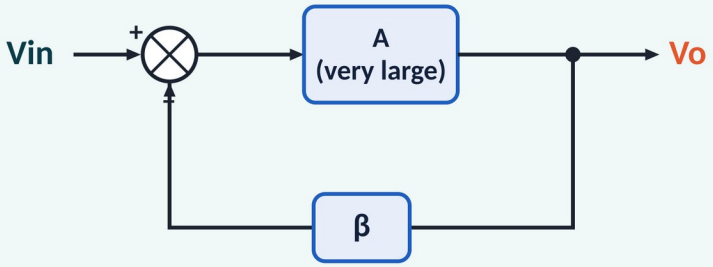
- State the ideal op-amp characteristics.
- Explain differential and common-mode operation and CMRR.
- Analyse inverting and non-inverting amplifiers.
- Design a summing amplifier.



Linked to CO5
Describe the working principles of op-amp and signal processing circuits.

Recap: What You Already Know

- Amplifier gain: A = Vo/Vin.
- Decibel: dB = 20 $\log_{10}$ A.
- Feedback: part of the output returned to the input.
- Kirchhoff's laws and Ohm's law.



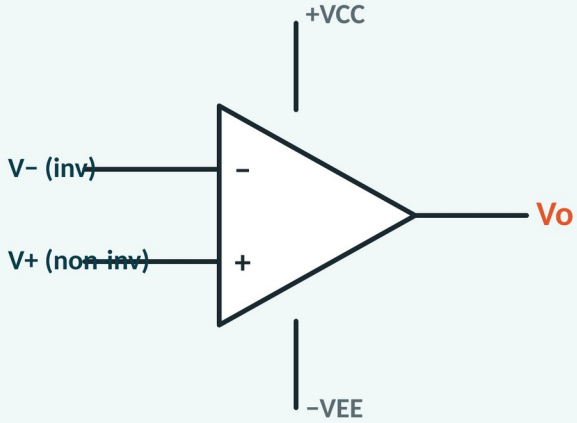
$V_o/V_{in} = A/(1 + A\beta) \approx 1/\beta$ (A very large)

virtual short: $V+ \approx V-$ and input current ≈ 0

Gain in dB = 20 $\log_{10}$ $|V_o/V_{in}|$; 40 dB = ×100

Motivation: The Universal Building Block

- Sensor amplifiers and audio equipment.
- Filters, oscillators, comparators.
- Analog computers: add, integrate, differentiate.
- One chip, many functions set by resistors.



$$V_o = A \cdot (V_+ - V_-)$$

Op-amp: a high-gain differential amplifier, usually with two supplies ±VCC

Op-Amp Symbol and Ideal Characteristics

- Two inputs: inverting (−) and non-inverting (+).
- One output; dual supply ±VCC.
- Ideal: infinite gain, infinite input resistance.
- Ideal: zero output resistance, infinite bandwidth.
- Ideal: infinite CMRR, zero offset.

Parameter	Ideal	Typical µA741
Open-loop gain A	∞	2×10^5
Input resistance	∞	2 MΩ
Output resistance	0	75 Ω
Bandwidth	∞	GBW 1 MHz
CMRR	∞	90 dB
Input offset	0	1–5 mV

$V_o = A \cdot (V+ - V-)$; practical µA741: $A \approx 2 \times 10^5$, $R_{in} \approx 2$ MΩ, $R_o \approx 75$ Ω

Op-Amp Specifications

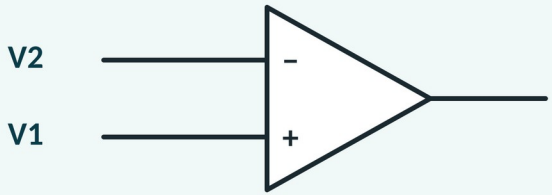
- Input offset voltage: output error with zero input.
- Input bias current: small base currents of input stage.
- Slew rate: limits output speed, V/µs.
- CMRR: ability to reject common signals.
- Gain–bandwidth product: A × f = constant.

Specification	µA741 (typical)	Meaning
Input offset voltage	2 mV	Vo ≠ 0 when Vin = 0
Input bias current	80 nA	base current of input stage
Slew rate	0.5 V/µs	max rate of change of Vo
CMRR	90 dB	|Ad/Ac| in decibels
Gain–bandwidth	1 MHz	A × f is constant
Output swing	±13 V	for ±15 V supply

Slew rate SR = dVo/dt|max (V/µs); fmax = SR/(2π·Vp) for an undistorted sine of peak Vp

Differential and Common-Mode Operation

- Differential signal: Vd = V1 − V2.
- Common-mode signal: Vc = (V1 + V2)/2.
- Op-amp amplifies Vd by Ad, Vc by Ac (small).
- CMRR measures how well Vc is rejected.
- High CMRR rejects noise picked up on both inputs.



Vd = V1 − V2

Vc = (V1 + V2)/2

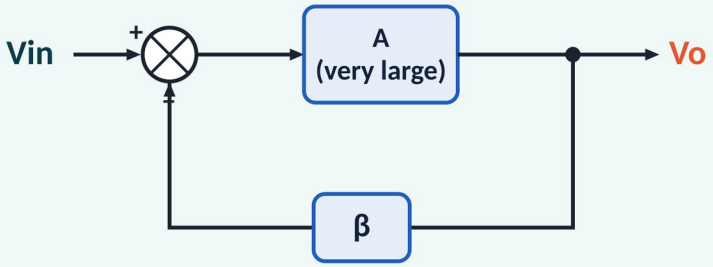
Vo = Ad·Vd + Ac·Vc

CMRR = |Ad/Ac|

$V_o = A_d \cdot V_d + A_c \cdot V_c$; $CMRR = |A_d/A_c|$; $CMRR(dB) = 20 \log_{10} |A_d/A_c|$

Negative Feedback and the Virtual Short

- Output fed back to the inverting input.
- Closed-loop gain ≈ 1/β, set by resistors.
- Because A is huge, V+ − V− ≈ 0: virtual short.
- Input current into the op-amp ≈ 0.
- Use these two rules for every circuit.



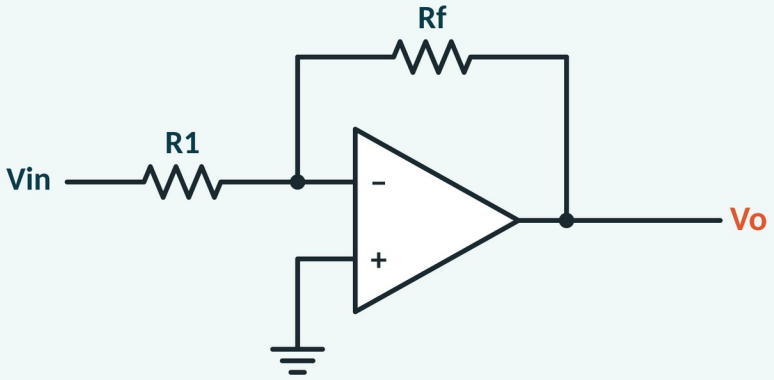
$V_o/V_{in} = A/(1 + A\beta) \approx 1/\beta$ (A very large)

virtual short: V+ ≈ V− and input current ≈ 0

$V_o/V_{in} = A/(1 + A\beta) \approx 1/\beta$; rules: $V+ = V-$, $I(in) = 0$

Inverting Amplifier

- Input applied through R1 to the (−) input.
- Non-inverting input grounded.
- Virtual ground at the (−) input.
- Current through R1 flows through Rf.
- Output is inverted.



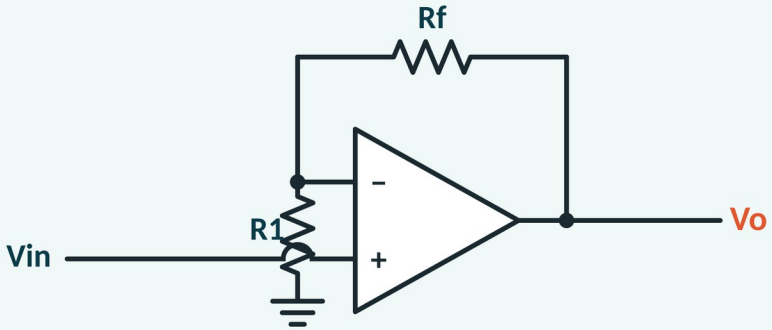
Vo = −(Rf/R1)·Vin

gain = −Rf/R1; input resistance = R1; Vo inverted

Vo = −(Rf/R1)·Vin; Av = −Rf/R1; Rin = R1

Non-Inverting Amplifier

- Input applied directly to the (+) input.
- R1 and Rf form a divider from the output.
- V− = Vo·R1/(R1 + Rf) = Vin.
- Gain is 1 + Rf/R1, at least 1.
- Very high input resistance.



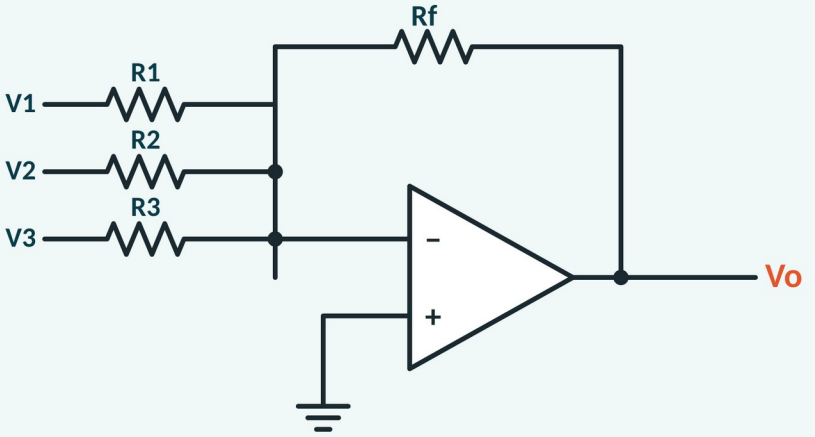
Vo = (1 + Rf/R1)·Vin

in phase; Rf = 0, R1 = ∞ → voltage follower (gain 1)

Vo = (1 + Rf/R1)·Vin; Av = 1 + Rf/R1
Voltage follower: Rf = 0, Av = 1

Summing Amplifier

- Several inputs, each through its own resistor.
- All join at the virtual ground.
- Currents add: I = V1/R1 + V2/R2 + V3/R3.
- Total current flows through Rf.
- Output is the inverted weighted sum.



Vo = −Rf(V1/R1 + V2/R2 + V3/R3)

equal resistors: Vo = −(V1 + V2 + V3)

Vo = −Rf·(V1/R1 + V2/R2 + V3/R3)

R1 = R2 = R3 = Rf: Vo = −(V1 + V2 + V3)

Solved Example 1: Inverting and Non-Inverting

- (a) Inverting: R1 = 10 kΩ, Rf = 100 kΩ, Vin = 0.2 V. (b) Non-inverting: R1 = 1 kΩ, Rf = 9 kΩ, Vin = 0.5 V.
- 1. Inverting gain = −Rf/R1 = −10.
- 2. Vo = −10 × 0.2 = −2 V; Rin = 10 kΩ.
- 3. Non-inverting gain = 1 + Rf/R1 = 1 + 9 = 10.
- 4. Vo = 10 × 0.5 = +5 V.

Inverting: Av = −Rf/R1 = −100k/10k = −10

Vo = −10 × 0.2 V = −2 V (within ±13 V)

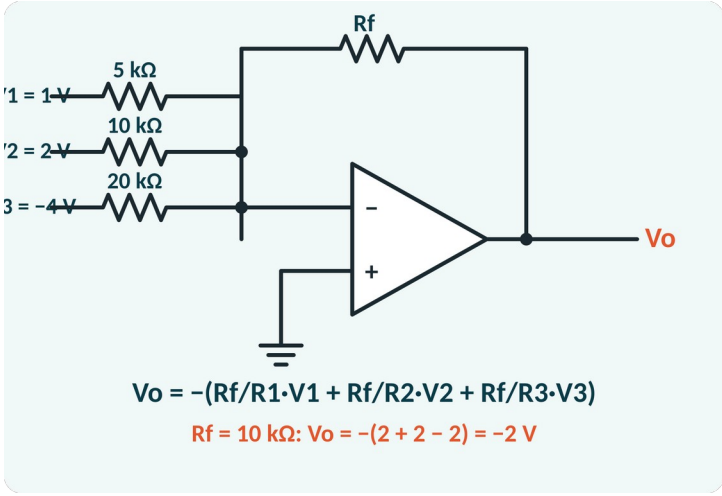
Non-inverting: Av = 1 + 9k/1k = 10

Vo = 10 × 0.5 V = +5 V; Rin(inv) = R1 = 10 kΩ

Answer: (a) Vo = −2 V; (b) Vo = +5 V; both within ±13 V swing

Solved Example 2: Summing Amplifier

- Rf = 10 kΩ, R1 = 5 kΩ, R2 = 10 kΩ, R3 = 20 kΩ; V1 = 1 V, V2 = 2 V, V3 = −4 V. Find Vo.
- 1. Weights: Rf/R1 = 2, Rf/R2 = 1, Rf/R3 = 0.5.
- 2. Sum = 2 × 1 + 1 × 2 + 0.5 × (−4) = 2 + 2 − 2 = 2.
- 3. Vo = −2 V.
- 4. Check: output within ±13 V.



Answer: Vo = −2 V

Common Mistakes and Exam Tips

Common mistakes

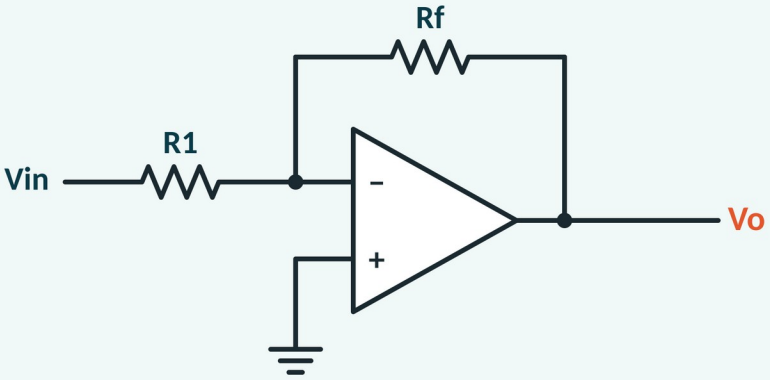
- Forgetting the sign in the inverting gain.
- Using 1 + Rf/R1 for the inverting amplifier.
- Ignoring op-amp saturation (±VCC).
- Treating Rin of the inverting amplifier as infinite.
- Applying virtual ground without negative feedback.

Exam tips

- Use V+ = V− and I(in) = 0.
- Check that |Vo| < supply voltage.
- Inverting: Rin = R1.
- Name the configuration before writing the formula.

Summary

- Ideal op-amp: A = ∞, Rin = ∞, Ro = 0, infinite CMRR.
- Differential gain Ad, common-mode gain Ac, CMRR = Ad/Ac.
- Negative feedback gives the virtual short.
- Inverting: −Rf/R1; non-inverting: 1 + Rf/R1.
- Summing: Vo = −Rf ΣVi/Ri.



$$V_o = -(R_f/R_1)\cdot V_{in}$$

gain = −Rf/R1

Rules: V+ = V−, no input current

Quick Quiz

Q1. Voltage gain of an inverting amplifier with R1 = 2 kΩ, Rf = 20 kΩ:
(a) 10 (b) −10 (c) 11 (d) −0.1

Ans: (b)

Q2. Non-inverting gain with Rf = 9 kΩ, R1 = 1 kΩ:
(a) 9 (b) 10 (c) −9 (d) 1

Ans: (b)

Q3. CMRR of 100 dB means |Ad/Ac| =
(a) 100 (b) 10^5 (c) 10^2 (d) 1000

Ans: (b)

Q4. Voltage follower has gain:
(a) 0 (b) 1 (c) 10 (d) ∞

Ans: (b)

Q5. At the inverting input of an inverting amplifier the voltage is:
(a) Vin (b) Vo (c) ≈ 0 (virtual ground) (d) VCC

Ans: (c)

References

1. **Boylestad Robert L. and Nashelsky Louis, Electronic Devices and Circuit Theory**
   11th Ed., PHI, 2016 (Units 3, 4, 5 textbook)

2. **David A. Bell, Operational Amplifier and Linear ICs**
   3rd Ed., Oxford University Press, 2011 (reference, Unit 5)

Next lecture: integrator, differentiator and comparator